CNENGR115 Computers and Control

Integral Action

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Control Lectures

Part 1 – Introduction ✓

Part 2 – Mathematical Models ✓

Part 3 – Control Actions

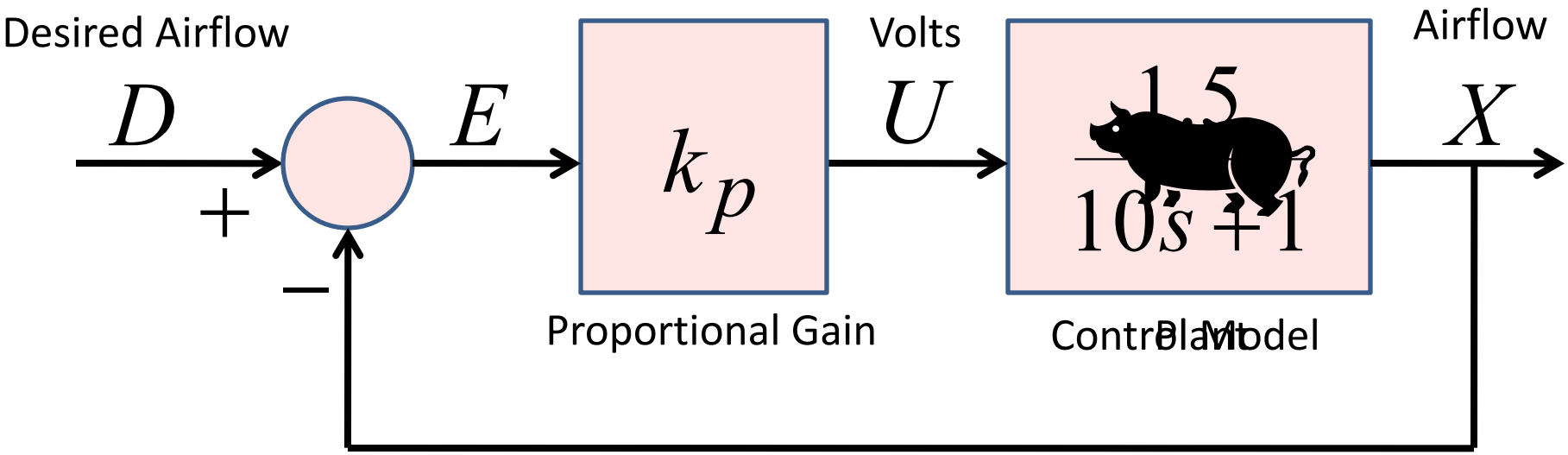
Proportional ✓

Integral

Proportional + Integral

Environmental Control

Proportional Control of Ventilation Rate (Recap)



The ventilation system includes disturbances, fan dynamics, dimensions of the room, sensors etc.

Proportional Control of Ventilation Rate (Recap)

Desired Airflow

$$D \longrightarrow \boxed{\frac{1.5k_p}{10s+1+1.5k_p}} \longrightarrow X$$

Airflow

"Closed–Loop Transfer Function"

Characteristic Equation

$$\tau s + 1 = 0$$

"Characteristic Equation of the generalised 1st order system"

$$10s + 1 = 0$$

"Ventilation system without control"

$$10s + 1 + 1.5k_p = 0$$

"Closed–loop control system"

$$\left(\frac{10}{1 + 1.5k_p}\right)s + 1 = 0$$

Re-arranged to determine the new time constant – we have **changed** the behaviour of this system!

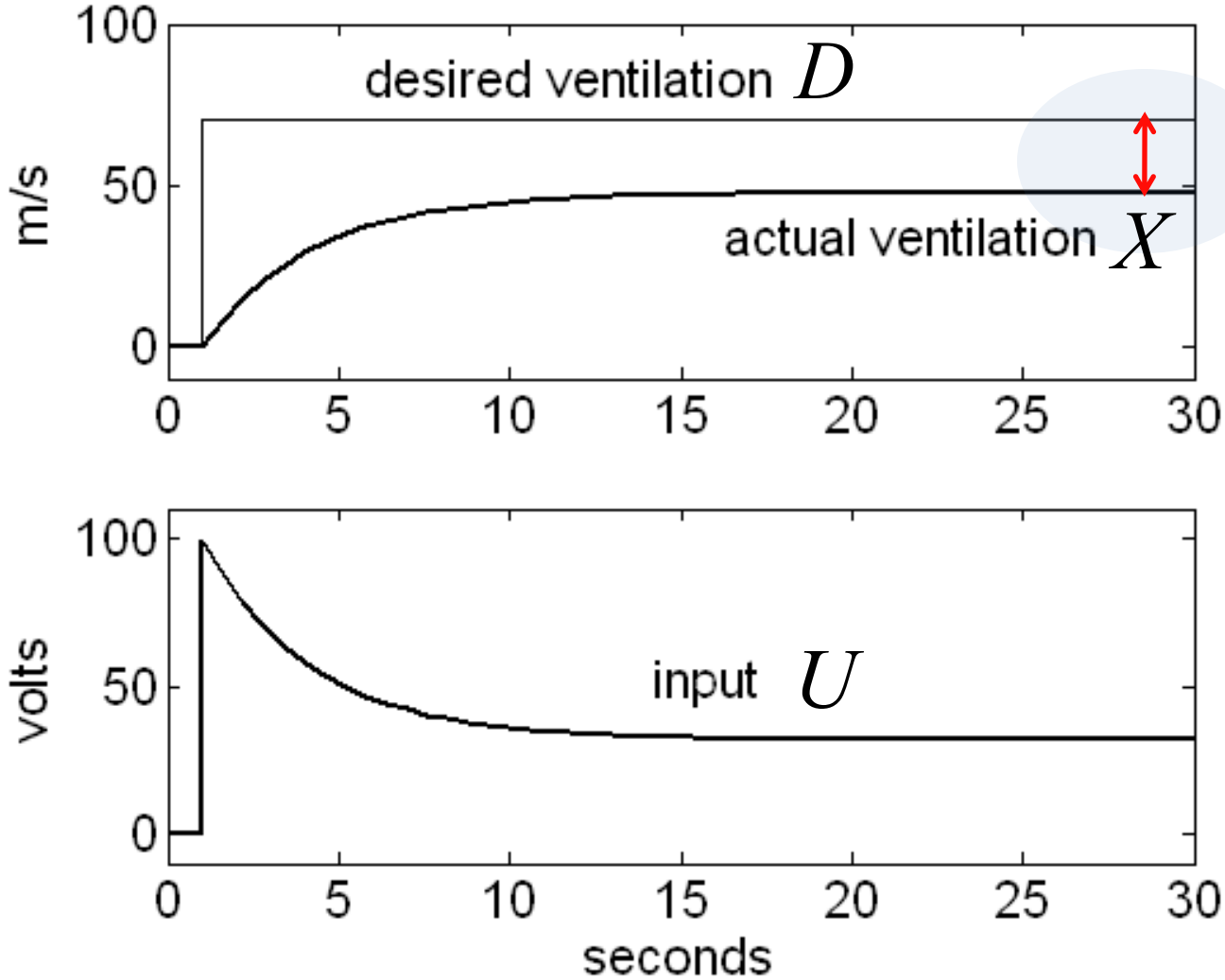
Steady State Gain

$$X = \frac{1.5k_p}{10s + 1 + 1.5k_p} D$$

$$x(t \to \infty) = \frac{1.5k_p}{1 + 1.5k_p} d$$

This is **less than unity** – at steady state, the actual ventilation rate will be less than the desired ventilation rate!

Proportional Control of Ventilation Rate



"Steady State Error"

Integral Control of Ventilation Rate

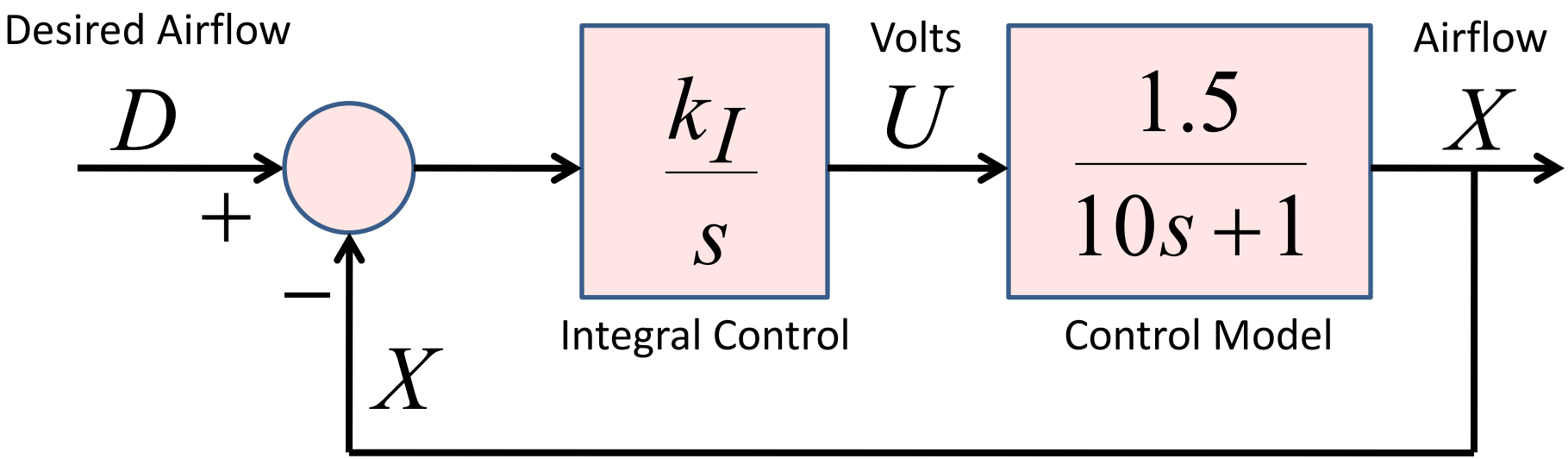
- An **integral control law** is given by

$$u = k_I \int (d - x)$$

$$U = \frac{k_I}{s}(D - X)$$

- At each time instant, the control input is the **integral of the error** signal multiplied by the control gain
- Block Diagram, Closed–Loop Transfer Function and Steady State Gain …

Integral Control of Ventilation Rate



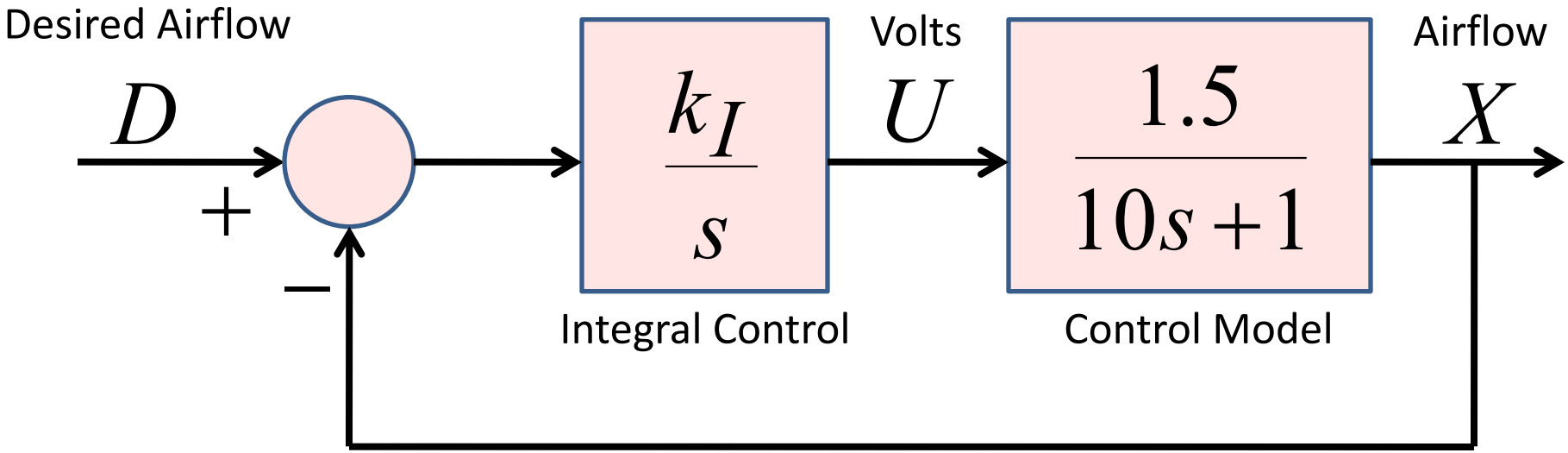
$$U = \frac{k_I}{s}(D - X) \qquad X = \frac{1.5}{10s+1}U$$

Exercise

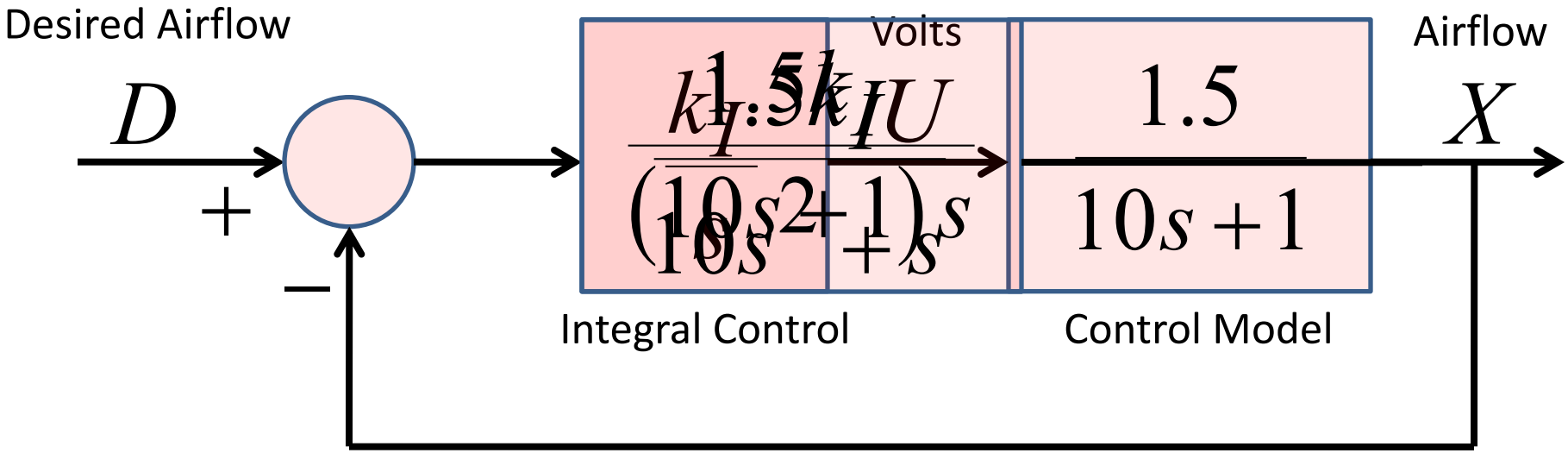
NOT AVAILABLE IN PREVIEW

$$u = k_p\left(d - \frac{dx}{dt}\right) \quad \rightarrow \quad U = k_p\left(D - sX\right)$$

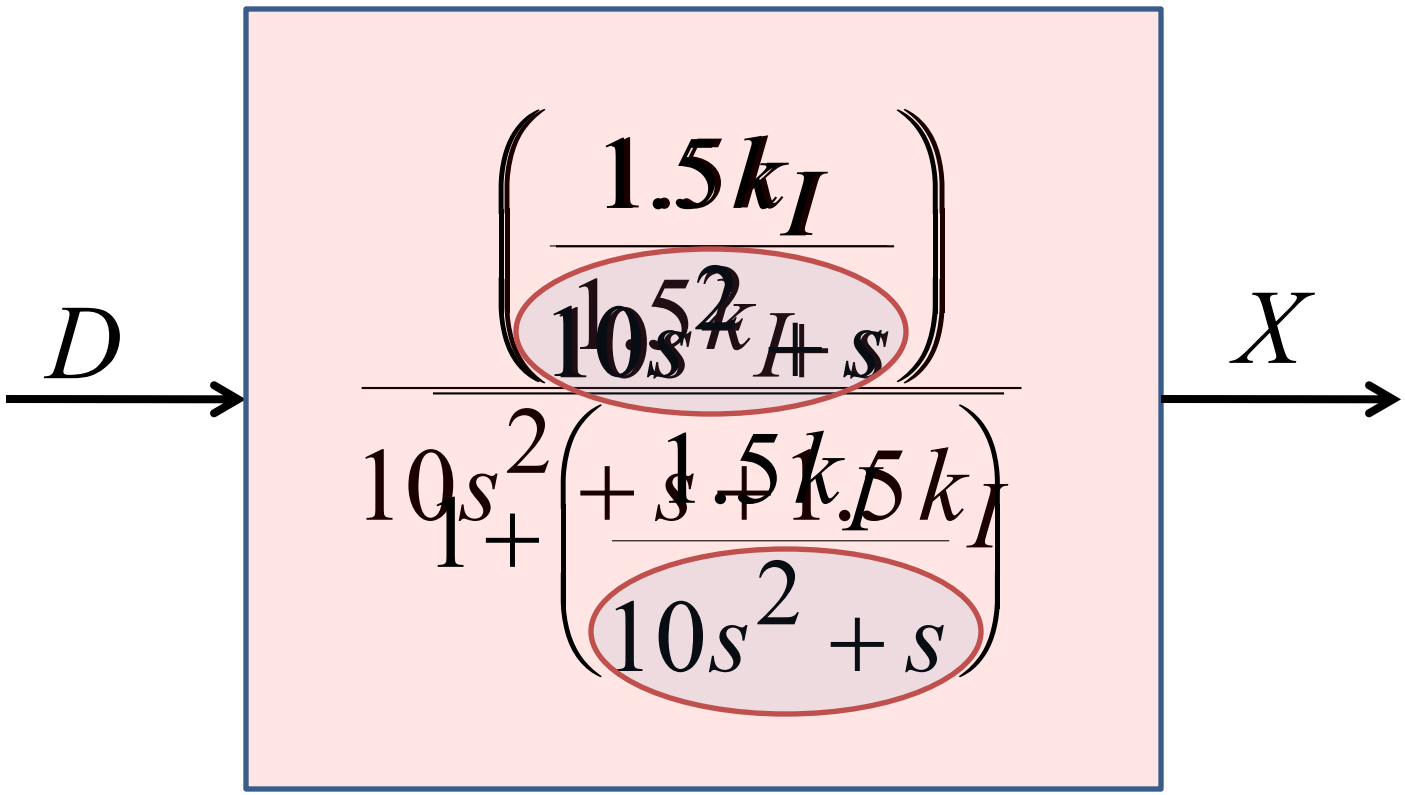
Integral Control of Ventilation Rate



Integral Control of Ventilation Rate



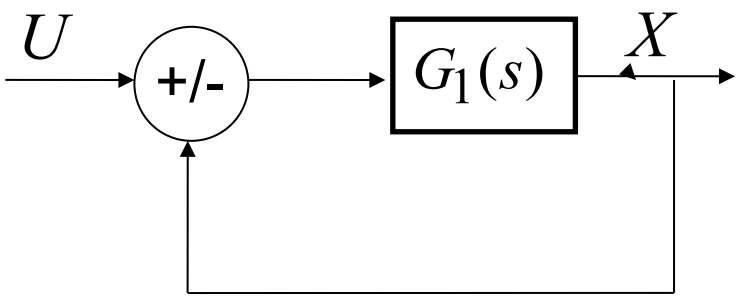
Integral Control of Ventilation Rate

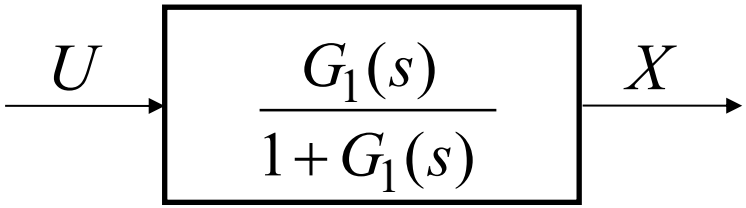


$$\frac{\left(\dfrac{1.5k_I}{10s^2+s}\right)}{1+\left(\dfrac{s+1.5k_I}{10s^2+s}\right)}$$

"Closed–Loop Transfer Function"

(Unity) Negative Feedback



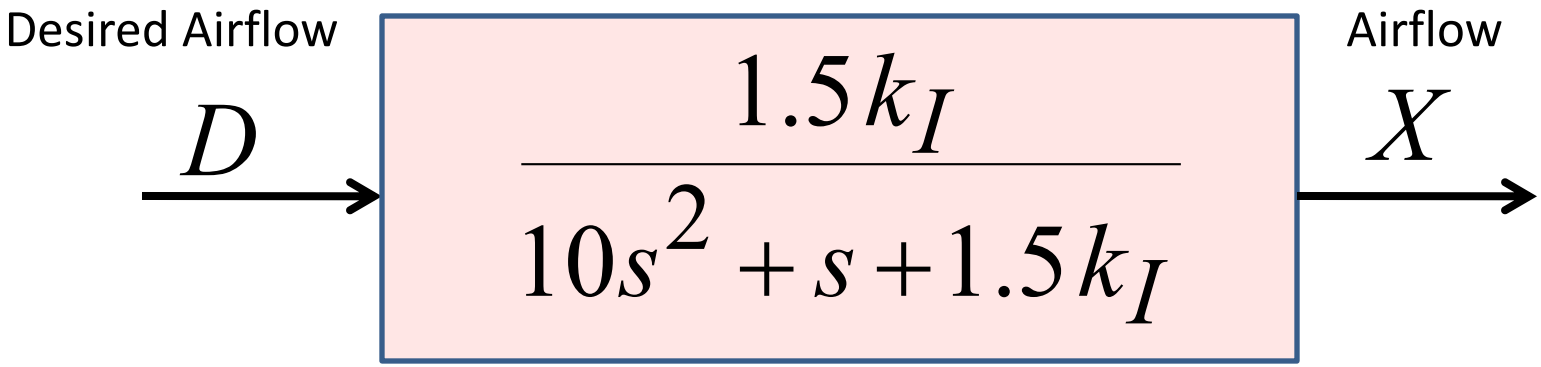


Closed–Loop Transfer Function

$$\frac{k_I}{s} \times \frac{1.5}{(10s+1)} = \frac{1.5k_I}{s(10s+1)}$$

$$X = \frac{\left(\dfrac{1.5k_I}{10s^2+s}\right)}{1+\left(\dfrac{1.5k_I}{10s^2+s}\right)} D = \frac{1.5k_I}{10s^2+s+1.5k_I} D$$

Integral Control of Ventilation Rate



"Closed–Loop Transfer Function"

$$X = \frac{1.5}{10s+1}U$$

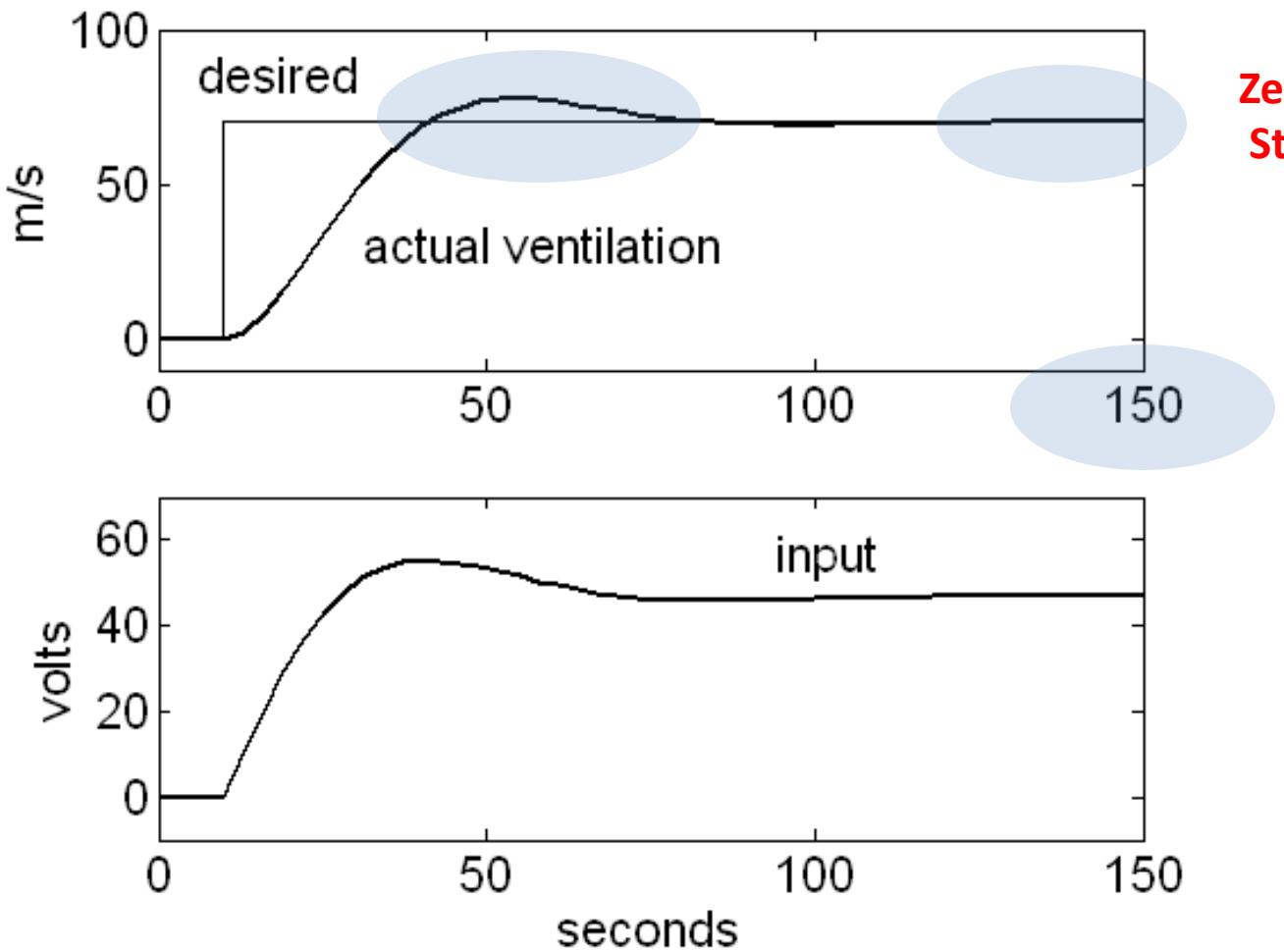
Steady State Gain for Integral Control

$$X = \frac{1.5k_I}{10s^2 + s + 1.5k_I} D$$

$$x(t \to \infty) = \frac{1.5k_I}{0 + 0 + 1.5k_I} D$$

The **steady state gain is unity** – at steady state, the actual ventilation rate will (asymptotically) approach the desired ventilation rate!

Integral Control of Ventilation Rate



Zero Steady State Error

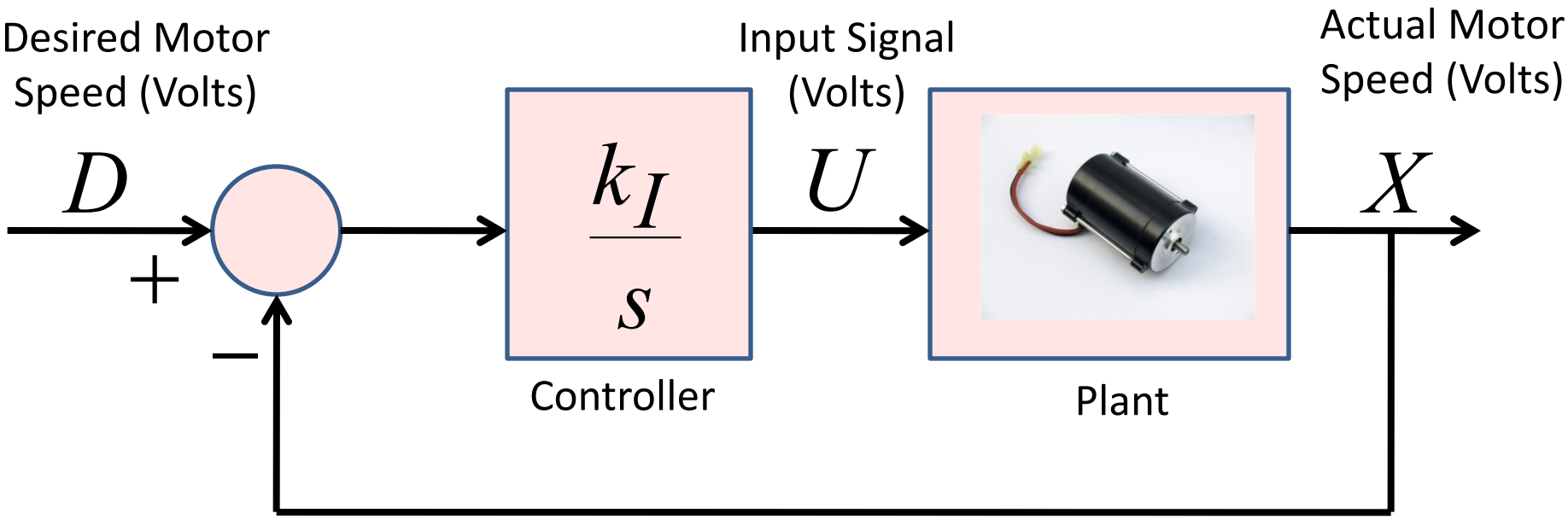
Steady State Gain for Integral Control

$$X = \frac{1.5k_I}{10s^2 + s + 1.5k_I} D$$

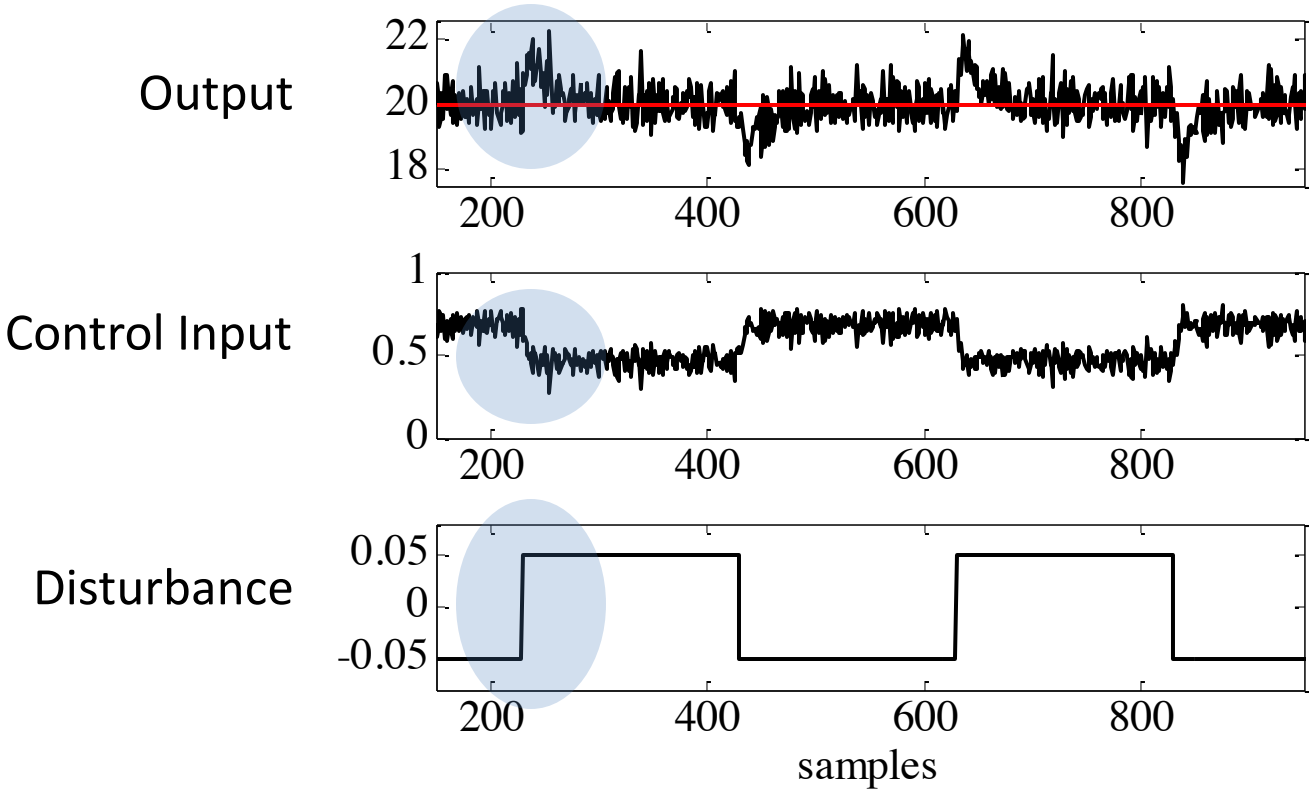
$$x(t \to \infty) = \frac{1.5k_I}{0 + 0 + 1.5k_I} D = D$$

Integral control yields steady state tracking of the set point (**even when there are modelling errors and disturbances**)

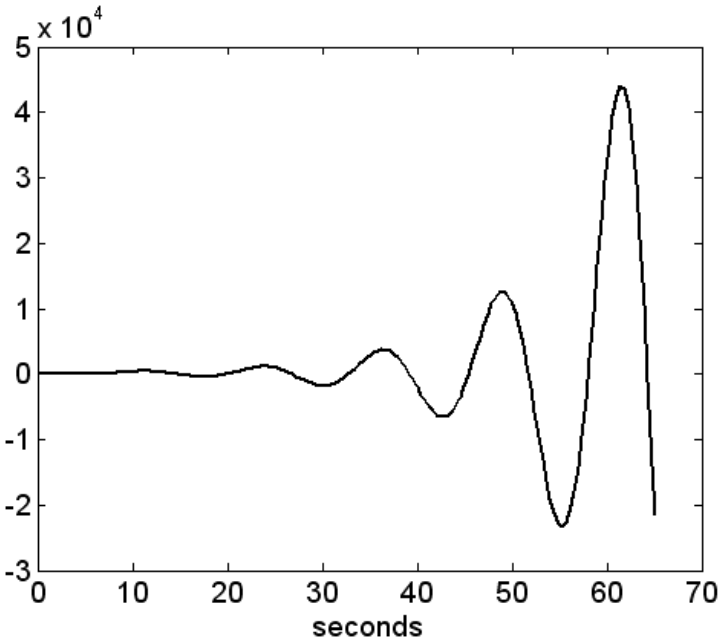
Integral Control of a DC Motor



Control of a DC Motor

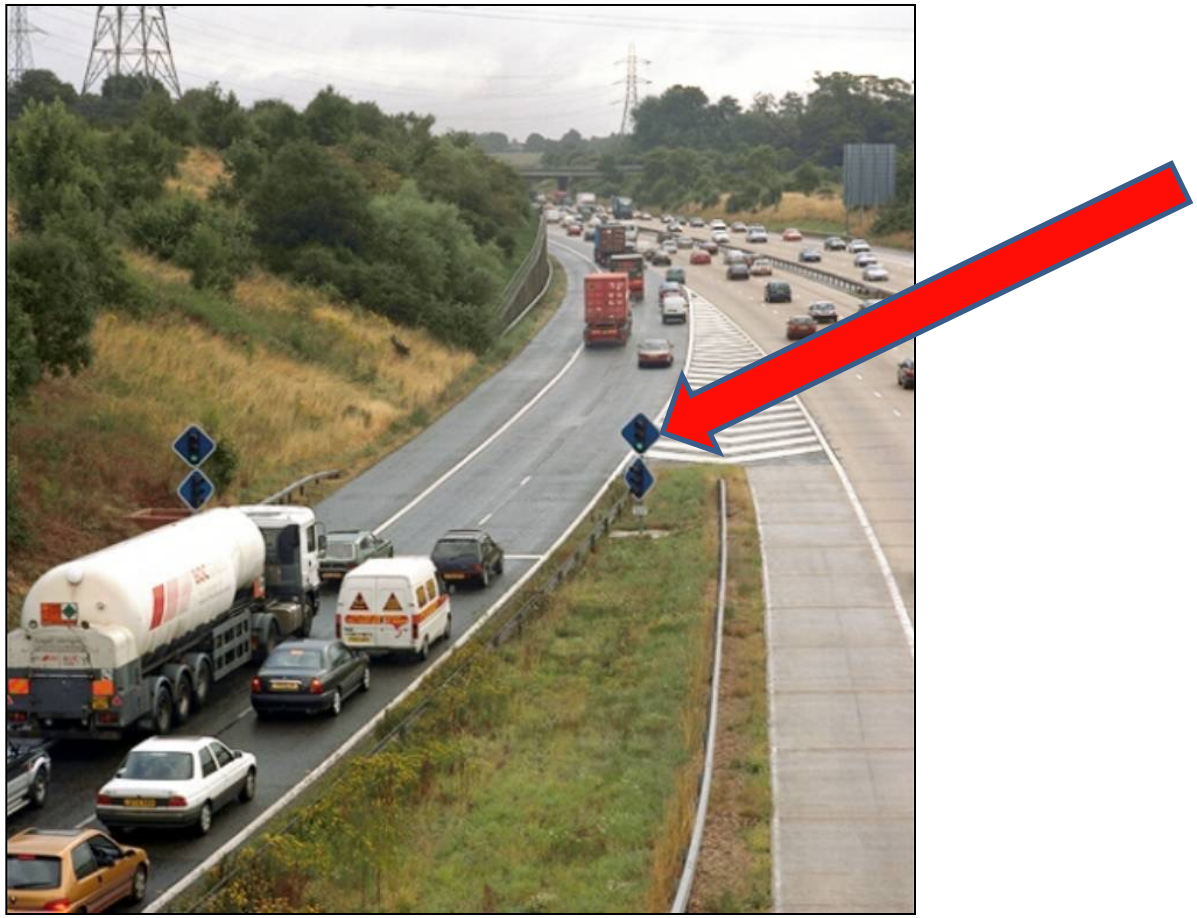


Quick warning …

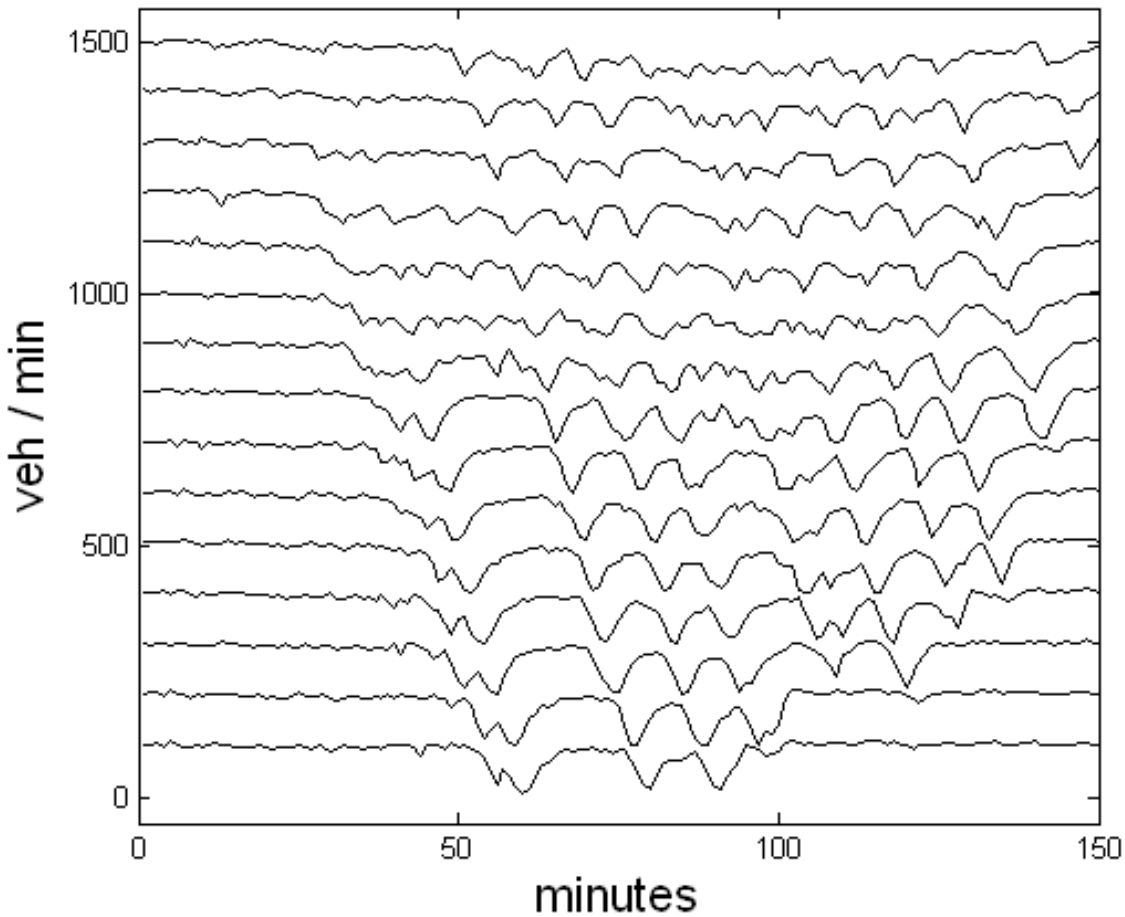


Integral control yields steady state tracking of the set point
(even when there are modelling errors and disturbances)
as long as the closed–loop system is stable

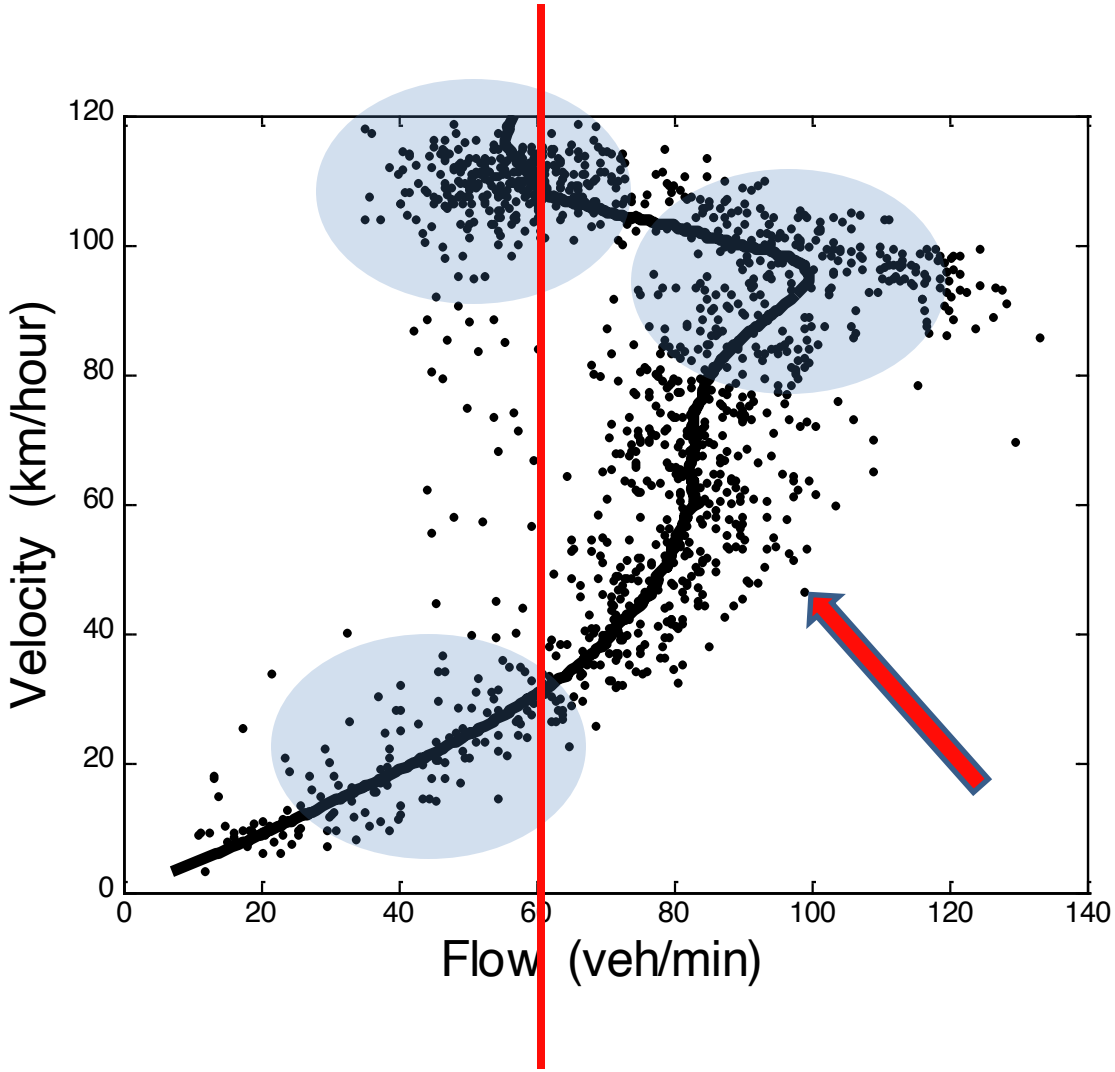
Ramp Metering Control on Motorways



Utrecht to Amsterdam A2 Motorway



M3 Motorway near Southampton



Ramp Metering Control on Motorways

$$U = \frac{k_I}{s} \cdot (D - X)$$

- Motorway **access control** uses traffic lights on the entrance ramps
- Vehicles temporarily delayed at the lights to avoid **flow breakdown** on the main carriageway of the motorway
- The industry standard algorithm (ALINEA) is based on integral control

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Proportional + Integral